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B51TF Assignment #1

Part A

A Combined Cycle Gas Turbine Plant (CCGT) includes a steam cycle operating at three pressures (Low, Intermediate and High) with steam being produced in a Heat Recovery Steam Generator (HRSG).

HRSG details

Wet saturated water at the output from the condenser is pumped to 2 bar and is passed through a feed water heater where it is heated to 100ºC.

(all pressures are absolute)

The feed is then split into three separate flows/pressures:

- 20% of the flow as a Low Pressure feed, 5.0bar, exiting as superheated steam at 200ºC
- 38% of the flow as a Intermediate Pressure feed, 36 bar, exiting as superheated steam at 350ºC
- The remainder of the flow as a High Pressure feed, 150 bar, exiting as superheated steam at 510oC

Each distinct pressure (feed to a turbine) has an economiser, evaporator and super heater. The HRSG receives 1000 kg/s of exhaust gases at 600ºC and has an acid dew point temperature of 105ºC and a design pinch point of 10 K.

Steam Circuit details

- High pressure feed enters the High Pressure (HP) turbine.
- Steam exiting the HP turbine **mixes** with the Intermediate Pressure (IP) feed before entering the IP turbine.
- Steam exiting the IP turbine **mixes** with the Low Pressure (LP) feed before entering the LP turbine.
- **Include one stage of reheating at a pressure and temperature of your choosing.**
- The LP turbine exhausts at the condenser pressure.
- The three turbines all have the same isentropic efficiency, 85%.
- The condenser operates at 0.07 bar.
- All pressures are absolute and there are no pressure losses in the circuit.

Determine;

- The outlet gas temperature from the HRSG [8 marks]
- The total mass flow rate of steam [2 marks]
- Net work output from the steam cycle [2 marks]

The specific heat capacity of the turbine exhaust gases is 1.15 kJ/kgK and may be assumed as constant. Steam properties can be found from steam tables or from [Steam Table Calculator | Spirax Sarco]

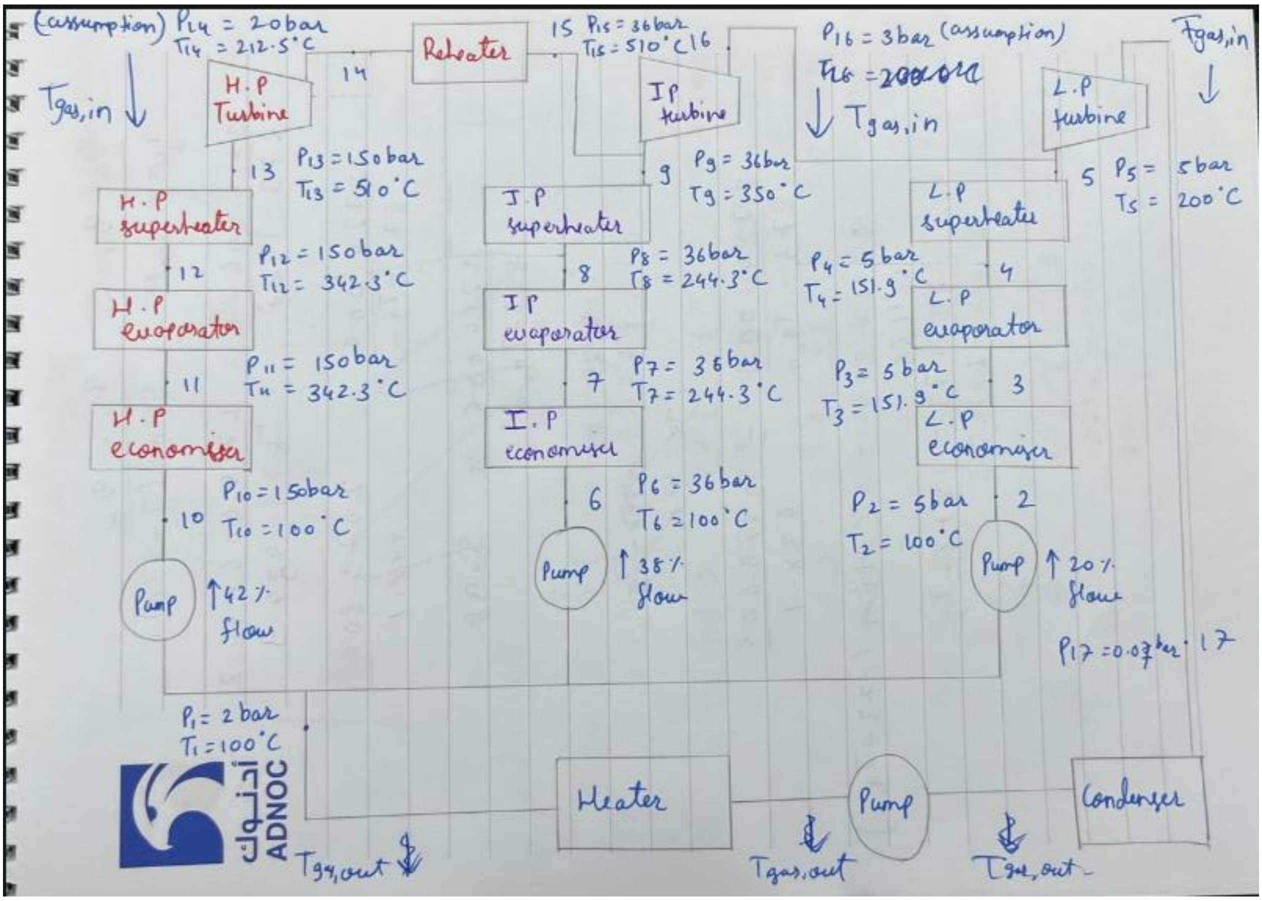
Solution:

CCGT plant has three pressure lines, Low, Intermediate and high pressure.

HRSG details:

- **Condenser:** The pressure is pumped at 2 bar, and temperature is raised to 100 C
- After condenser feed is then split into three separate flows:
  - Low pressure: 20 % of flow, exiting at 200C (superheated), P = 5 bar
  - Intermediate pressure: 38 % flow , exiting at T 350 C (superheated) , P = 36 bar
  - High pressure: 42% flow, exiting at T 510 C (superheated) , P = 150 bar
- Each pressure line has economiser, evaporator and super heater
- Mass flow rate of hot gas: 1000 kg/s , at T = 600 C ,
- Dew point, T = 105 C
- Three turbines have efficiency of 85 %

Circuit Diagram:



Assumption:

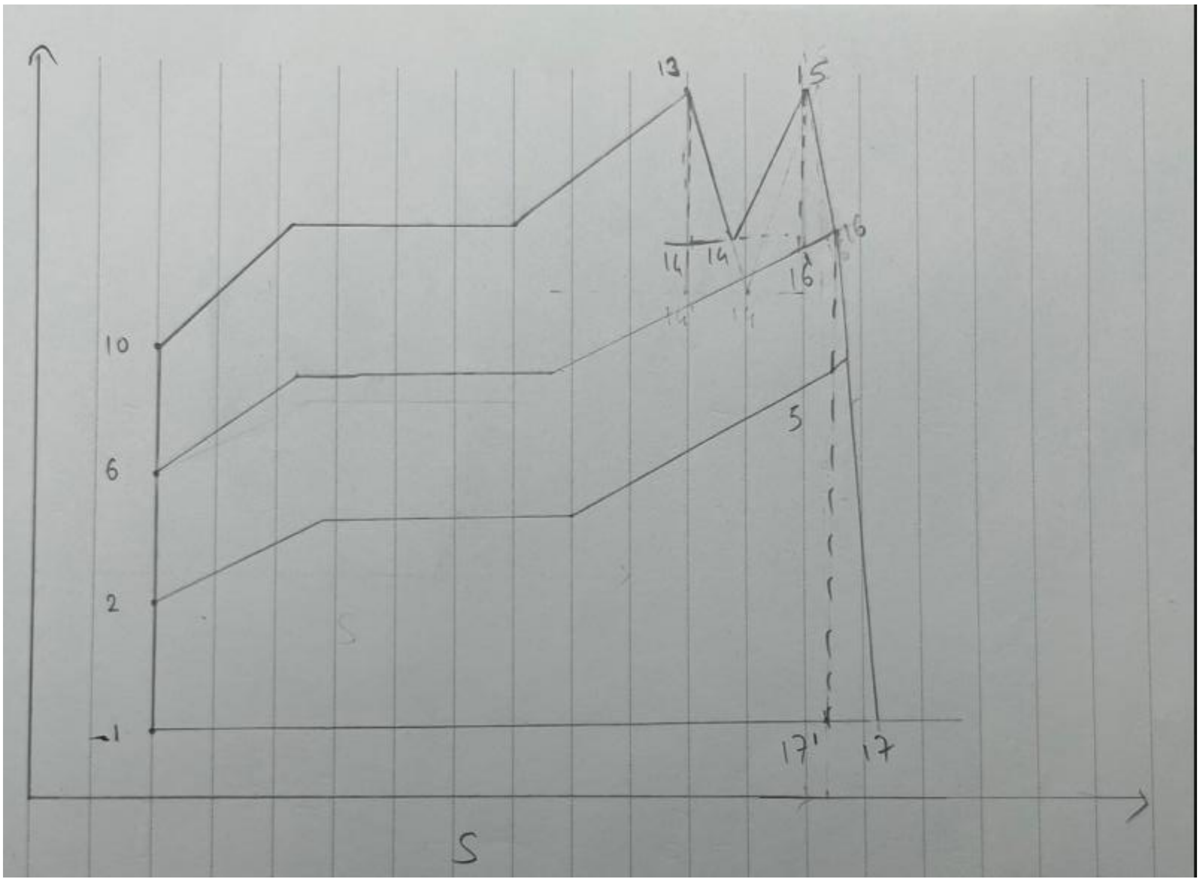
T input reheater: 485.5 K, pressure: 20 bar

T output reheater: 783 K, pressure: 36 bar

Intermediate turbine: outlet P= 3 bar

Low pressure turbine: outlet P = 0.07 bar

TS diagram:



Calculation:

For calculation I am just showing all pressure lines values in tabulated form and calculation are in Appendix:

Calculation: (Enthalpy and Q table)

- **Low pressure**

Unit	**Temperature (K)**	**Enthalpy (kJ/kg)**	**Q (kJ/kg)**
Low Pressure Economiser	T_{in} = 373 T_{out} = 424.9	h_{in} = 419.4 h_{out} = 640.4	Q_{in} = 0 Q_{out} = 0 + 0.2 * (640.4-419.4) = 44.2
Low Pressure Evaporator	T_{in} = 424.9 T_{out} = 424.9	h_{in} = 640.4 h_{out} = 2749	Q_{in} = 44.2 Q_{out} = 44.2 + 0.2 * (2749-640.4) = 465.92
Low Pressure Super heater	T_{in} = 424.9 T_{out} = 473	h_{in} = 2749 h_{out} = 2855	Q_{in} = 465.92 Q_{out} = 465.92 + 0.2 * (2855 - 2749) = 487.12

- **Intermediate pressure**

Unit	**Temperature (K)**	**Enthalpy (kJ/kg)**	**Q (kJ/kg)**
Intermediate Pressure Economiser	T_{in} = 373 T_{out} = 517.3	h_{in} = 422.8 h_{out} = 1058	Q_{in} = 487.12 Q_{out} = 487.12 + 0.38 * (1058-422.8) = 728.49
Intermediate Pressure Evaporator	T_{in} = 517.3 T_{out} = 517.3	h_{in} = 1058 h_{out} = 2803	Q_{in} = 728.49 Q_{out} = 728.49 + 0.38 * (2803-1058) = 1391.59
Intermediate Pressure Super heater	T_{in} = 517.3 T_{out} = 623	h_{in} = 2803 h_{out} = 3101	Q_{in} = 1391.59 Q_{out} = 1391.59 + 0.38 * (3101 - 2803) = 1504.83

- **High Pressure**

Unit	**Temperature (K)**	**Enthalpy (kJ/kg)**	**Q (kJ/kg)**
High Pressure Economiser	T_{in} = 373 T_{out} = 615.3	h_{in} = 434.2 h_{out} = 1611	Q_{in} = 1504.83 Q_{out} = 1504.83 + 0.42 * (1611-434.2) = 1999.086
High Pressure Evaporator	T_{in} = 615.3 T_{out} = 615.3	h_{in} = 1611 h_{out} = 2611	Q_{in} = 1999.086 Q_{out} = 1999.086 + 0.42 * (2611-1611) = 2419.086
High Pressure Super heater	T_{in} = 615.3 T_{out} = 783	h_{in} = 2611 h_{out} = 3337	Q_{in} = 2419.086 Q_{out} = 2419.086 + 0.42 * (3337-2611) = 2724.006
Re-heater	T_{in} = 485.5 T_{out} = 783	h_{in} = 2745 h_{out} = 3472.4	Q_{in} = 2724.006 Q_{out} = 2724.006+ 0.42 * (3472.4-2745) = 3029.514

Re-heater enthalpy calculation (Appendix 4)

Now we have to find the pinch point, by gradient method: the point which has least gradient will be selected as coordinate for triangular method to find $T_{gas,out}$.

After doing calculation coordinate (Q,T): (44.2,424.9) is pinch point and after adding 10 K to it, the coordinate becomes (44.2,434.9), gas coordinate: (3029.51, 873)

Slope with minimum values was = 0.142 **(Appendix 5)**

Solving for $T_{gas,out}$: triangle similarity method

$$(Q_4 – Q_1) / (T_{g,in} – T_{g,out}) = (Q_4 – Q_{pinch}) / T_{h,in} – (T_2 + \Delta T_{pinch})$$

$$(3029.51-0) / (873 – T_{g,out}) = (3029.51 – 44.2) / (873 – 434.9)$$

$$\mathbf{T_{g,out} = 428.4123\ K}$$

Solving for mass flow rate of steam:

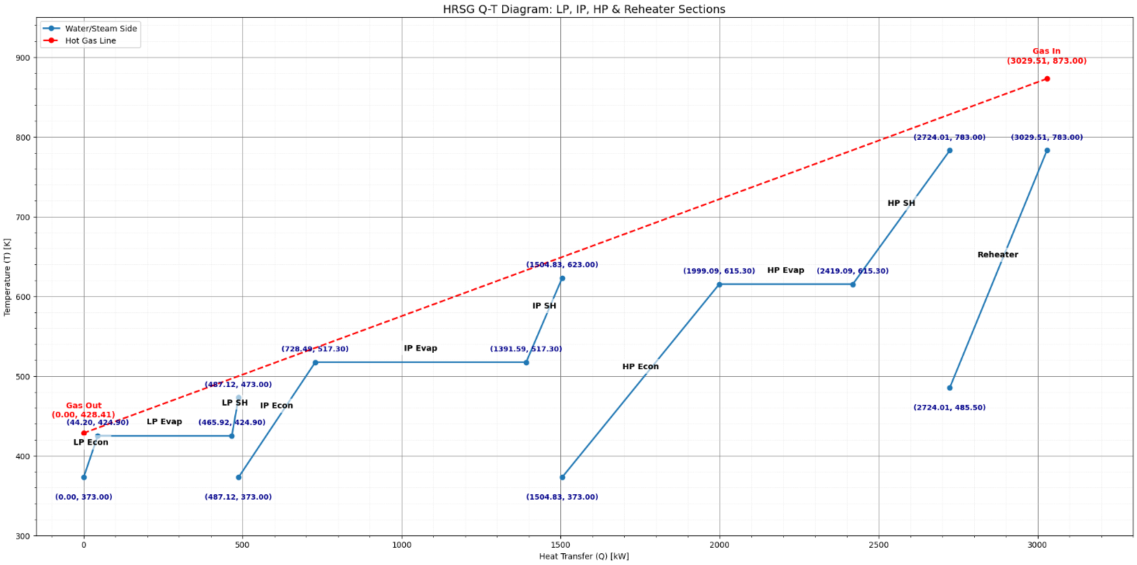
$$m_s = [\ 1000(kg/s) * 1.15\ (kJ/kg\ K) * (\ 873 – 428.4123)\ ] / [3472.4 – 419.4\ ]$$

$$\mathbf{m_s = 167.466\ kg/s}$$

Net Work output: (Appendix 6)

$$W_{net} = W_{out} – W_{in}$$
$$W_{net} = 731.03-19.29$$
$$\mathbf{W_{net} = 711.71\ kW}$$

TQ Diagram:


HRSG Q-T Diagram: LP, IP, HP & Reheater Sections
Water/Steam Side
Hot Gas Line
Temperature (T) [K]
Heat Transfer (Q) [kW]
Gas In
(3029.51, 873.00)
Gas Out
(0.00, 428.41)
(0.00, 373.00)
LP Econ
(44.20, 424.90)
LP Evap
(465.92, 424.90)
LP SH
(487.12, 473.00)
(487.12, 373.00)
IP Econ
(728.49, 517.30)
IP Evap
(1391.59, 517.30)
IP SH
(1504.83, 623.00)
(1504.83, 373.00)
HP Econ
(1999.09, 615.30)
HP Evap
(2419.09, 615.30)
HP SH
(2724.01, 783.00)
Reheater
(2724.01, 485.50)
(3029.51, 783.00)

Part B

The following process streams are available in a process plant.

Stream Number	Type	t_i (°C)	t_t(°C)	m (kg/s)	c_p (J/kg°K)
1	hot	169.2	30.5	5.44	4414
2	hot	144.0	31.5	1.45	13126
3	hot	111.5	70.9	22.59	7351
4	hot	70.9	45.0	4.62	2532
5	hot	62.8	60.4	11.84	135717#
6	hot	114.0	27.8	0.6	2215
7	hot	173.5	149.2	26.98	2586
8	hot	173.5	49.1	1.4	2209
9	hot	440.0	279.5	10.92	1222
10	cold	38.4	147.6	7.19	2161
11	cold	230.7	234.1	119.95	22616
12	cold	175.7	176.6	72.06	103146
13	cold	209.4	218.0	11.99	3554
14	cold	90.1	91.1	21.17	58007
15	cold	114.1	122.4	50.37	8802
16	cold	-28.4	20.0	0.9	8512
17	cold	77.0	250.0	0.83	14972

large specific heat capacity over a small temperature range = change of phase

Calculate the plant cooling and heating requirements based on a minimum pinch value of 12 K [8 marks]

Solution:
To calculate Q total for heating I use python code, the code is in appendix,
Initially I arranged hot and cold stream data into numpy array, and calculated the value using formula: **Q = m*Cp* ΔT** for hot and cold stream and then added it:

Q_c_total = 25404.6 kW
Q_h_total = 20709.6 kW

Heading utility calculation: (Appendix 7)

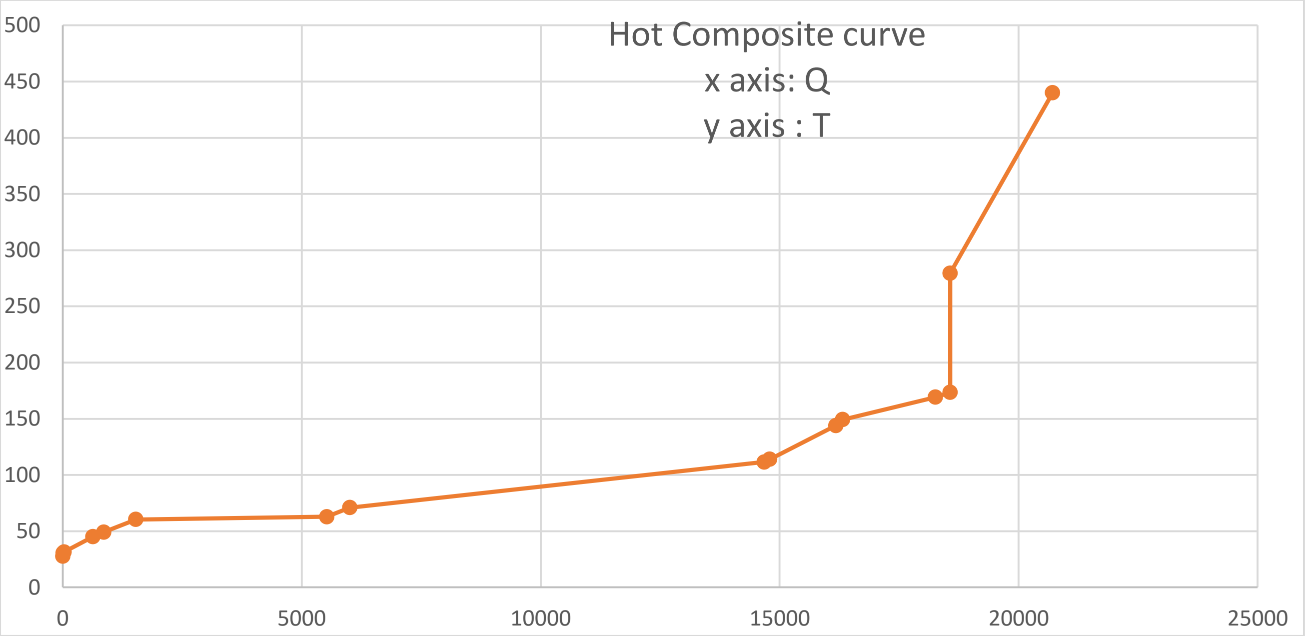
```
For T_initial = 169.2 , T_final= 30.5 , Mass = 5.44 , Specific heat = 4414
The value of Q is: 3330486.592 W

For T_initial = 144.0 , T_final= 31.5 , Mass = 1.45 , Specific heat = 13126
The value of Q is: 2141178.75 W

For T_initial = 111.5 , T_final= 70.9 , Mass = 22.59 , Specific heat = 7351
The value of Q is: 6741999.053999999 W

For T_initial = 70.9 , T_final= 45.0 , Mass = 4.62 , Specific heat = 2532
The value of Q is: 302974.0560000001 W

For T_initial = 62.8 , T_final= 60.4 , Mass = 11.84 , Specific heat = 135717
The value of Q is: 3856534.2719999976 W

For T_initial = 114.0 , T_final= 27.8 , Mass = 0.6 , Specific heat = 2215
The value of Q is: 114559.8 W

For T_initial = 173.5 , T_final= 149.2 , Mass = 26.98 , Specific heat = 2586
The value of Q is: 1695417.8040000007 W

For T_initial = 173.5 , T_final= 49.1 , Mass = 1.4 , Specific heat = 2209
The value of Q is: 384719.44 W

For T_initial = 440.0 , T_final= 279.5 , Mass = 10.92 , Specific heat = 1222
The value of Q is: 2141750.52 W

For heating utility total Q_hot_utility: 20709.6 kW
```

Cooling utility calculation: (Appendix 8)

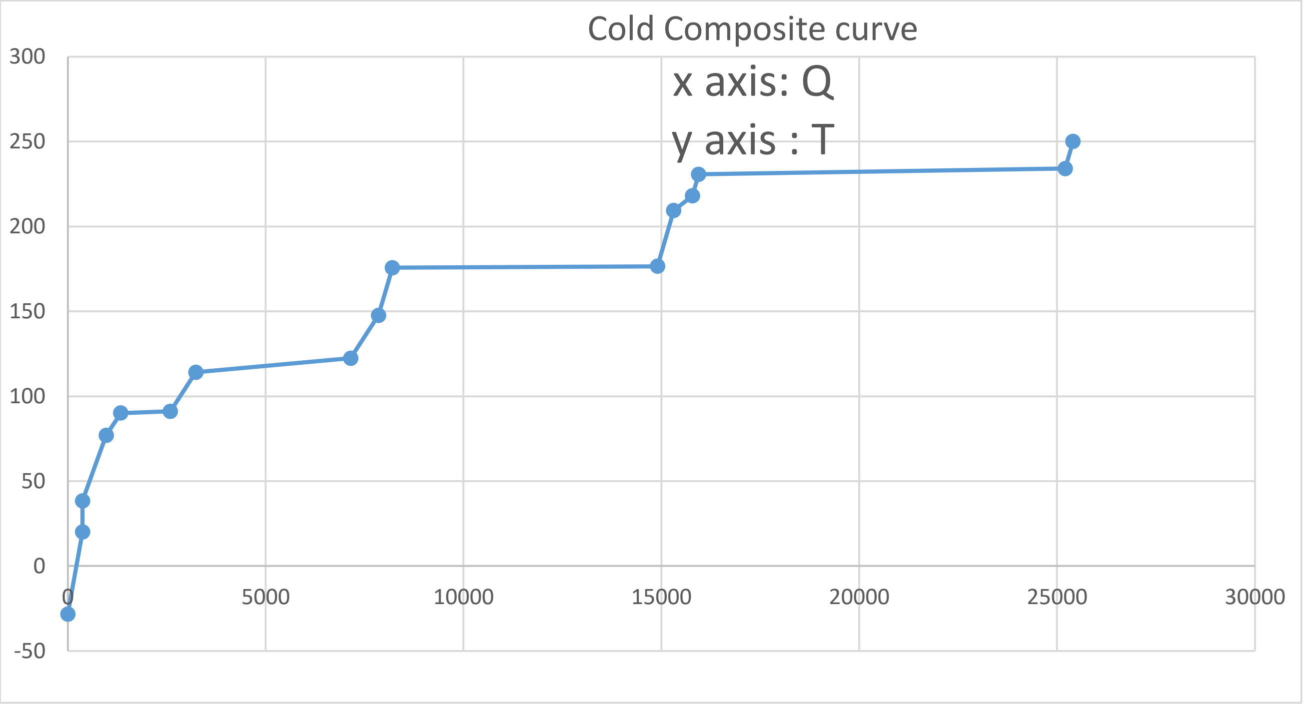
```
For T_initial = 38.4 , T_final= 147.6 , Mass = 7.19 , Specific heat = 2161
The value of Q is: 1696704.8279999997 W

For T_initial = 230.7 , T_final= 234.1 , Mass = 119.95 , Specific heat = 22616
The value of Q is: 9223483.280000016 W

For T_initial = 175.7 , T_final= 176.6 , Mass = 72.06 , Specific heat = 103146
The value of Q is: 6689430.684000042 W

For T_initial = 209.4 , T_final= 218.0 , Mass = 11.99 , Specific heat = 3554
The value of Q is: 366467.1559999997 W

For T_initial = 90.1 , T_final= 91.1 , Mass = 21.17 , Specific heat = 58007
The value of Q is: 1228008.1900000002 W

For T_initial = 114.1 , T_final= 122.4 , Mass = 50.37 , Specific heat = 8802
The value of Q is: 3679860.942000005 W

For T_initial = -28.4 , T_final= 20.0 , Mass = 0.9 , Specific heat = 8512
The value of Q is: 370782.72 W

For T_initial = 77.0 , T_final= 250.0 , Mass = 0.83 , Specific heat = 14972
The value of Q is: 2149829.48 W

For cooling utility Q_cool_utility: 25404.6 kW
```

After utility calculation, we find temperature intervals for each composite curve, from lowest to highest, to identify which stream falls into which interval. This calculations are done in excel sheet, I am just pasting composite curve in this word document. The calculation excel sheet is emailed to Dr Stephen.

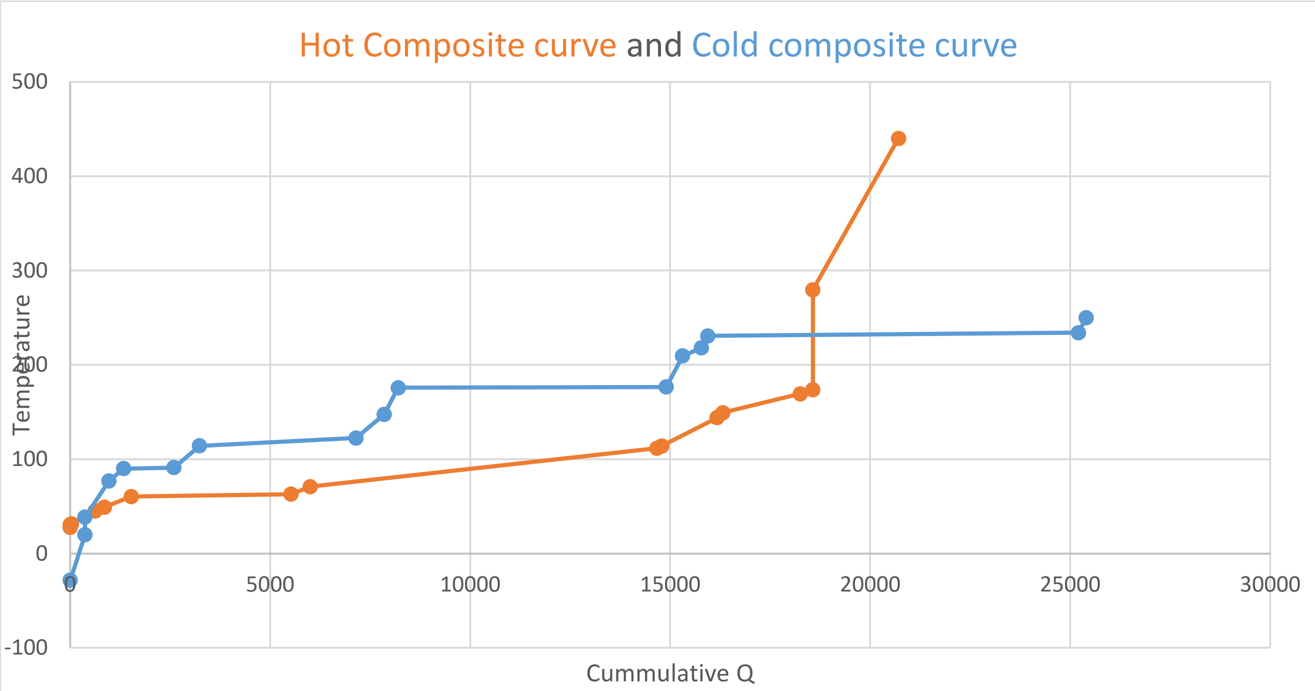
Hot Composite curve:



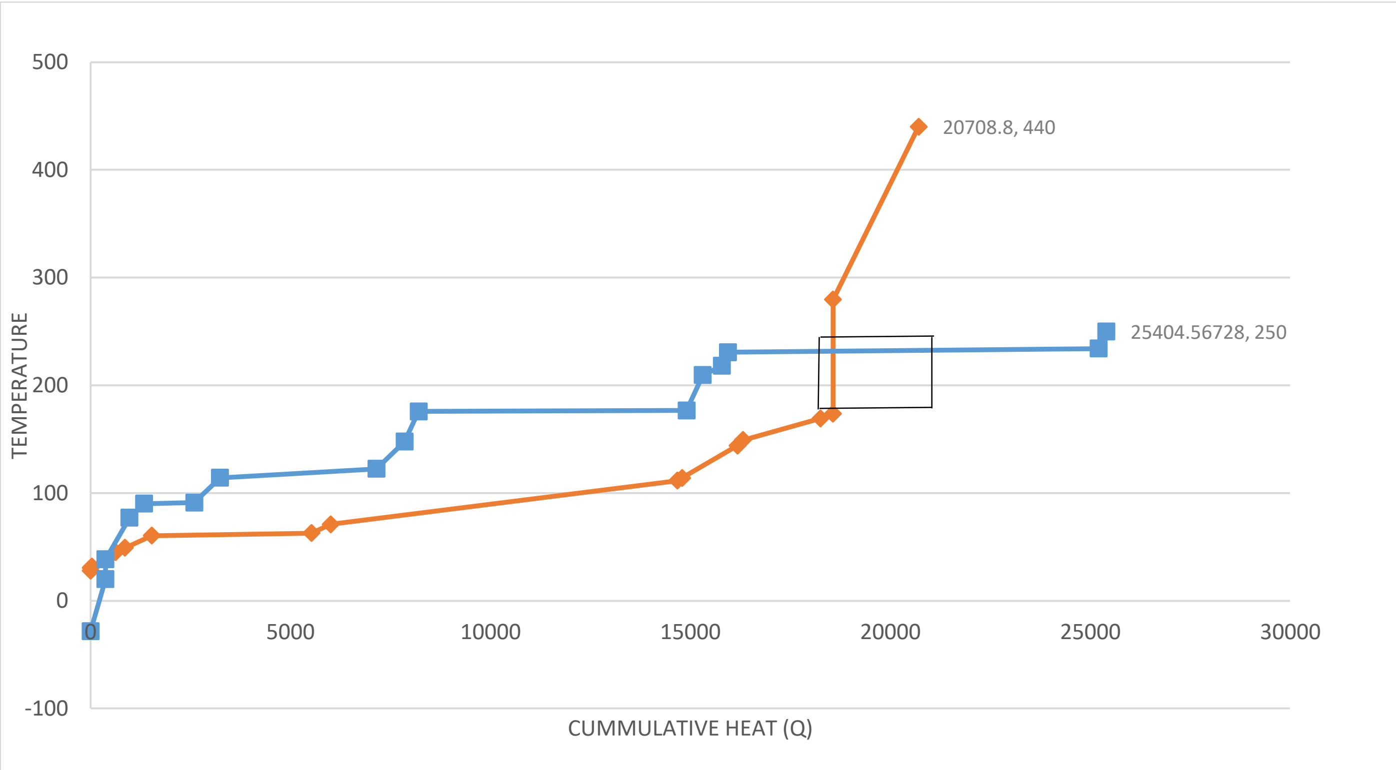
Cold composite curve:



Hot and Cold composite curve:



Violation curve:



The intersection in black box, is greatest violation as hot line can never cross cold line

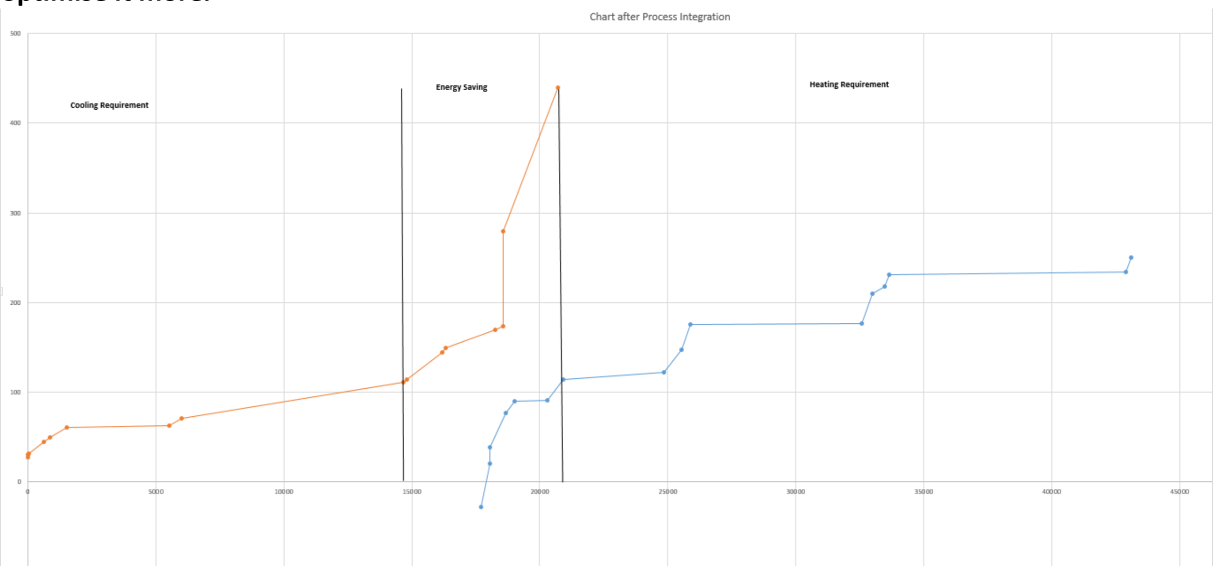
So now we have to find **Q offset,** (Check Excel Sheet for detailed calculation)

To find **Q offset** I found the slope at coordinates:
(Q,T): (18254.4335,169.2) , (18567.7401,173.5)

Note: At this coordinate x2 – x1 is zero so used points above:
(Q,T): (18567.74005,173.5) , (18567.7401,279.5)

So,
Slope m is 0.013724575
Intercept c is -81.33434506
Equation of line: T = 0.01372Q-81.3343

At point, temperature T = 173.5 K , after using delta T pinch of 12K or 12 C we get: T = 161.5 C
Substituting in equation of line: Q = 17693.39605
Coordinate: (17693.39605, 161.5)
Looking at graph below, the temperature difference is not 12 K at any point so decided to optimise it more.



When we use,
T = 161.5 C, in equation **T = 0.01372Q - 81.3343** ;
Extra Heating requirement: 43097.96333 – 20929.41532 = 22168.54801 kW
Extra Cooling requirement: 17693.39605- 0 = 17693.39605 kW

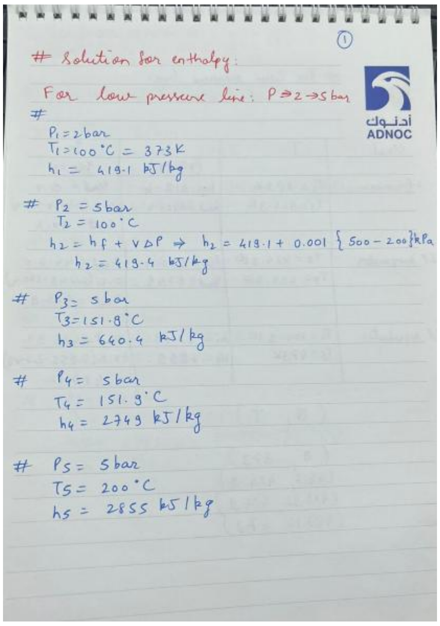
Using coordinate (85, 12119.45305) we get graph below: (it satisfies the 12 K pinch point)
And energy saving is maximised.

T = 85 C, in equation **T = 0.01372Q - 81.3343** ;
Extra heating requirement: 37524.02033 – 20900 = 16604.02033 kW
Extra cooling requirement: 12119.45305 – 0 = 12119.45305 kW

Marking Guidance (not a rubric)		
Part	**Description**	**marks**
	In both sections students should consider a simple and concise presentation; all repetitive calculation do not need to be shown, given an exemplar calculation of each 'type' and then present the values in a table. Clear diagrams are expected.	
A	Steam Circuit analysis with appropriate T-s diagram and circuit layout	4
	Construction of T-Q diagram for steam circuit and HRSG noting sensible and considered arrangement of components	4
	Determination of Mass flow rate of steam	2
	Determination of Net work output from the steam cycle	2
B	Process integration; completion of heating and cooling tables; completion of appropriate T-Q combined stream curves and identification additional heating and cooling requirements	2 4 2

Appendix:

Appendix 1: Low pressure



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Solution for enthalpy:

For low pressure line: P ⇒ 2 → 5 bar

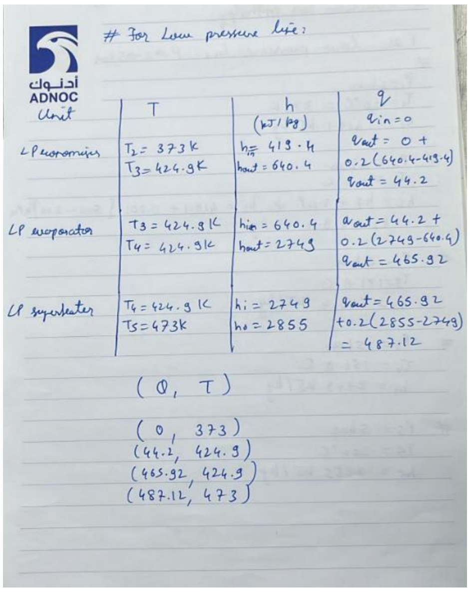
#
$P_1 = 2$ bar
$T_1 = 100\,^\circ C = 373 K$
$h_1 = 419.1$ kJ/kg

$P_2 = 5$ bar
$T_2 = 100\,^\circ C$
$h_2 = h_f + v\Delta P \Rightarrow h_2 = 419.1 + 0.001\{500 - 200\}$ kPa
$h_2 = 419.4$ kJ/kg

$P_3 = 5$ bar
$T_3 = 151.9\,^\circ C$
$h_3 = 640.4$ kJ/kg

$P_4 = 5$ bar
$T_4 = 151.9\,^\circ C$
$h_4 = 2749$ kJ/kg

$P_5 = 5$ bar
$T_5 = 200\,^\circ C$
$h_5 = 2855$ kJ/kg



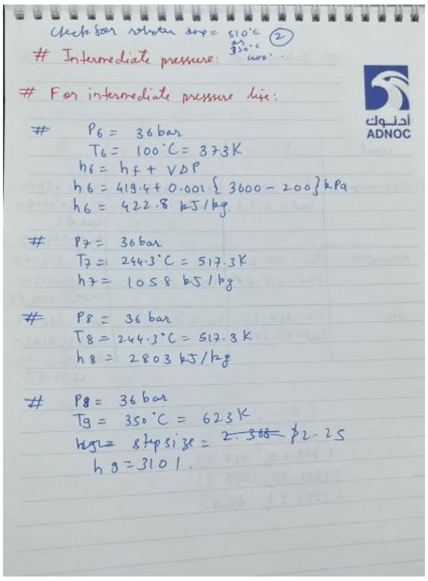
For Low pressure line:

Unit	T	h (kJ/kg)	q
			$q_{in} = 0$
LP economizer	$T_2 = 373 K$ $T_3 = 424.9 K$	$h_{in} = 419.4$ $h_{out} = 640.4$	$q_{out} = 0 + 0.2(640.4 - 419.4)$ $q_{out} = 44.2$
LP evaporator	$T_3 = 424.9 K$ $T_4 = 424.9 K$	$h_{in} = 640.4$ $h_{out} = 2749$	$q_{out} = 44.2 + 0.2(2749 - 640.4)$ $q_{out} = 465.92$
LP superheater	$T_4 = 424.9 K$ $T_5 = 473 K$	$h_i = 2749$ $h_o = 2855$	$q_{out} = 465.92 + 0.2(2855 - 2749)$ $= 487.12$

(Q, T)

(0, 373)
(44.2, 424.9)
(465.92, 424.9)
(487.12, 473)

Appendix 2: Intermediate pressure



Check for solution step = 510 °C or 350 °C 2

Intermediate pressure:

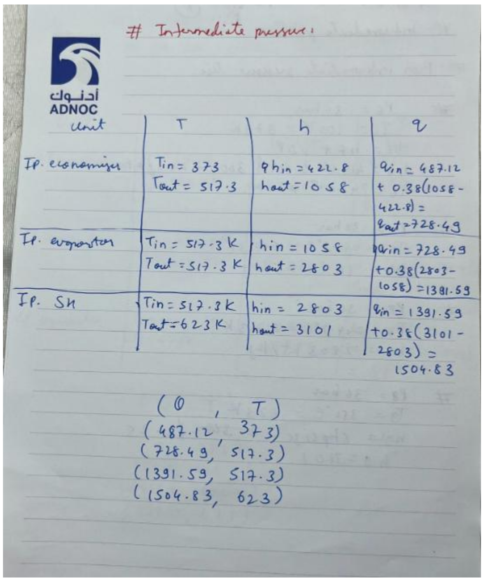
For intermediate pressure line:

$P_6 = 36$ bar
$T_6 = 100\,^\circ C = 373 K$
$h_6 = h_f + v\Delta P$
$h_6 = 419.4 + 0.001\{3600 - 200\}$ kPa
$h_6 = 422.8$ kJ/kg

$P_7 = 36$ bar
$T_7 = 244.3\,^\circ C = 517.3 K$
$h_7 = 1058$ kJ/kg

$P_8 = 36$ bar
$T_8 = 244.3\,^\circ C = 517.3 K$
$h_8 = 2803$ kJ/kg

$P_8 = 36$ bar
$T_9 = 350\,^\circ C = 623 K$
h_9 = stepsize = ~~2.368~~ 2.25
$h_9 = 3101$.

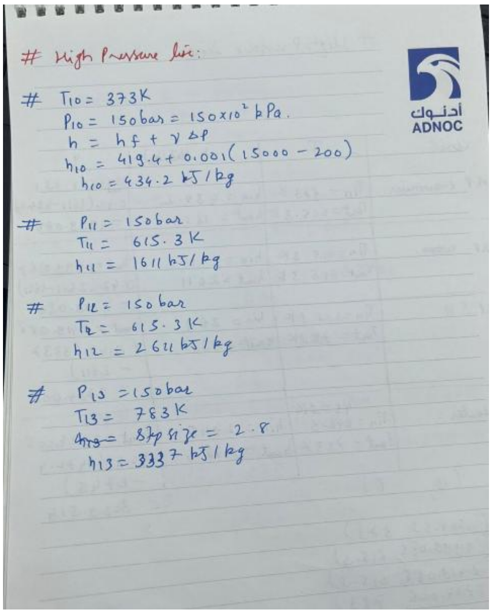


Intermediate pressure:

Unit	T	h	q
IP. economizer	$T_{in} = 373$ $T_{out} = 517.3$	$h_{in} = 422.8$ $h_{out} = 1058$	$q_{in} = 487.12 + 0.38(1058 - 422.8) =$ $q_{out} = 728.49$
IP. evaporator	$T_{in} = 517.3 K$ $T_{out} = 517.3 K$	$h_{in} = 1058$ $h_{out} = 2803$	$q_{in} = 728.49 + 0.38(2803 - 1058) = 1391.59$
IP. SH	$T_{in} = 517.3 K$ $T_{out} = 623 K$	$h_{in} = 2803$ $h_{out} = 3101$	$q_{in} = 1391.59 + 0.38(3101 - 2803) =$ 1504.83

(Q, T)
(487.12, 373)
(728.49, 517.3)
(1391.59, 517.3)
(1504.83, 623)

Appendix 3: High pressure



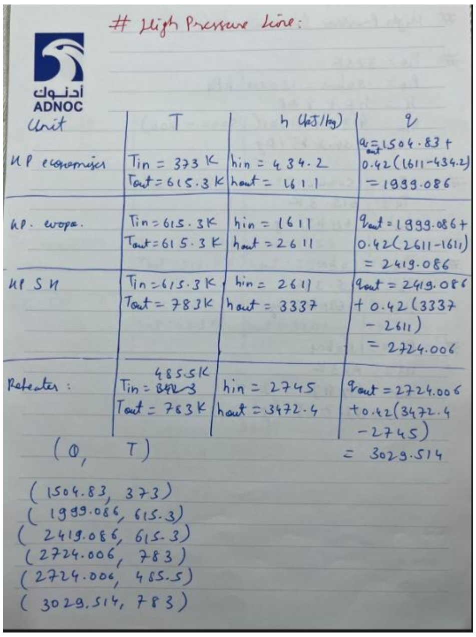
\# High Pressure line:

\# $T_{10} = 373K$
$P_{10} = 150bar = 150\times10^2 kPa.$
$h = h_f + v\Delta P$
$h_{10} = 419.4 + 0.001(15000 - 200)$
$h_{10} = 434.2 \; kJ/kg$

\# $P_{11} = 150bar$
$T_{11} = 615.3K$
$h_{11} = 1611 \; kJ/kg$

\# $P_{12} = 150bar$
$T_{12} = 615.3K$
$h_{12} = 2611 \; kJ/kg$

\# $P_{13} = 150bar$
$T_{13} = 783K$
Step size = 2.8
$h_{13} = 3337 \; kJ/kg$



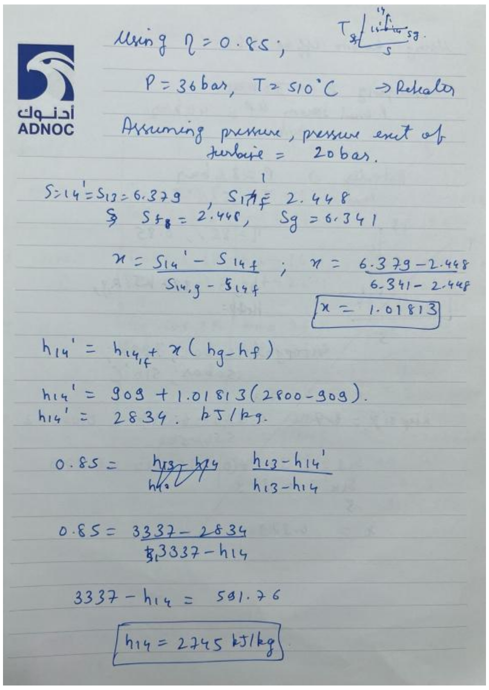
\# High Pressure Line:

Unit	T	h (kJ/kg)	q
HP economiser	$T_{in} = 373K$ $T_{out} = 615.3K$	$h_{in} = 434.2$ $h_{out} = 1611$	$q_{out} = 1504.83 + 0.42(1611-434.2) = 1999.086$
HP. evapo.	$T_{in} = 615.3K$ $T_{out} = 615.3K$	$h_{in} = 1611$ $h_{out} = 2611$	$q_{out} = 1999.086 + 0.42(2611-1611) = 2419.086$
HP SH	$T_{in} = 615.3K$ $T_{out} = 783K$	$h_{in} = 2611$ $h_{out} = 3337$	$q_{out} = 2419.086 + 0.42(3337 - 2611) = 2724.006$
Reheater:	$T_{in} = 485.5K$ $T_{out} = 783K$	$h_{in} = 2745$ $h_{out} = 3472.4$	$q_{out} = 2724.006 + 0.42(3472.4 - 2745) = 3029.514$

(Q, T)

(1504.83, 373)
(1999.086, 615.3)
(2419.086, 615.3)
(2724.006, 783)
(2724.006, 485.5)
(3029.514, 783)

Appendix 4: Re-heater enthalpy calculation



Using $\eta = 0.85$;

$P = 36bar, \; T = 510°C$ → Reheater

Assuming pressure, pressure exit of turbine = 20bar.

$s_{14}' = s_{13} = 6.379$, $s_{14f} = 2.448$
$s_{fg} = 2.448$, $s_g = 6.341$

$x = \frac{s_{14}' - s_{14f}}{s_{14,g} - s_{14f}}$; $x = \frac{6.379 - 2.448}{6.341 - 2.448}$

$\boxed{x = 1.01813}$

$h_{14}' = h_{14,f} + x(h_g - h_f)$

$h_{14}' = 909 + 1.01813(2800 - 909)$.
$h_{14}' = 2834. \; kJ/kg$.

$0.85 = \frac{h_{13} - h_{14}'}{h_{13} - h_{14}}$

$0.85 = \frac{3337 - 2834}{3337 - h_{14}}$

$3337 - h_{14} = 591.76$

$\boxed{h_{14} = 2745 \; kJ/kg}$

Appendix 5: (Selection of T_{pinch} point)

Gasline : (3029.51, 873)

$\Delta T_P \Rightarrow 10K$: 8

① → (2724.006, 783)

$$grad = \frac{873 - (783 + 10)}{3029.51 - 2724.006} = 0.261$$

② → (1999.086, 615.3)

$$grad = \frac{873 - (615.3 + 10)}{3029.51 - 1999.086} = 0.2403$$

③ → (1504.83, 623)

$$grad = \frac{873 - (623 + 10)}{3029.51 - 1504.83} = 0.1639$$

④ ⇒ (728.49, 517.3)

$$grad = \frac{873 - (517.3 + 10)}{3029.51 - 728.49} = 0.1502$$

⑤ ⇒ (487.12, 473)

$$grad = \frac{873 - (473 + 10)}{3029.51 - 487.12} = 0.1533$$

⑥ → (44.2, 424.9)

$$grad = \frac{873 - (424.9 + 10)}{3029.51 - 44.2} = 0.146$$

Appendix 6: ($W_{net\ calculation}$)

~~Wnet = ṁ (hout - hin~~

$$\dot{W}_{out} = \dot{m}\,(h_{in} - h_{out})_{HP}$$
$$= 0.42\,(3337 - 2611)$$
$$= 0.42 \times 1\,(304.9)$$
$$= 128\ kJ/s = 128\ kW$$

$$\dot{W}_{out} = \dot{m}\,(h_{in} - h_{out})_{IP}$$
$$= \cancel{(3235.85}$$
$$= \dot{m}\,(3295.985 - 3235.84)$$
$$= (0.38 + 0.42)\,(3295.955 - 3235.84)$$
$$= 48.11\ kJ/kg = 48.11\ kW$$

$$\dot{W}_{out} = \dot{m}\,(h_{in} - h_{out})_{LP}$$
$$= 1\,\dot{m}\,(3159.67 - 2604.75)_{LP}$$
$$= 554.92\ kW$$

$\dot{W}_{HP} = 128\ kW$, $\dot{W}_{IP} = 48.11\ kW$, $\dot{W}_{LP} = 554.92\ kW$

$$W_{in} = W_{pump} + W_{pump,HP} + W_{pump,IP} + W_{pump,LP}$$

$$W_{in} = (419.293 - 419.1) + (434.2 - 419.1) + (422.8 - 419.1) + (419.4 - 419.1)$$

$$W_{in} = 0.193 + 15.1 + 3.7 + 0.3$$

$$\boxed{W_{in} = 19.29\ kW}$$

$$W_{net} = W_{out} - W_{in}$$
$$= (128 + 48.11 + 554.92) - (19.29)$$

$$W_{net} = 731.03 - 19.29$$
$$\boxed{W_{net} = 711.74\ kW}$$

Appendix 7: (Heating utility python code)

```
import numpy as np

T_initial_hot = np.array([169.2, 144.0, 111.5, 70.9, 62.8, 114.0, 173.5, 173.5, 440.0])

T_initial_cold = np.array([30.5, 31.5, 70.9, 45.0, 60.4, 27.8, 149.2, 49.1, 279.5])

m = np.array([5.44, 1.45, 22.59, 4.62, 11.84, 0.6, 26.98, 1.4, 10.92])

Cp = np.array([4414, 13126, 7351, 2532, 135717, 2215, 2586, 2209, 1222])

z = 0

def q_utility(T_h, T_c, mass, spec_heat):

    return mass * spec_heat * (T_h - T_c)

for h, c, mass, cap in zip(T_initial_hot, T_initial_cold, m, Cp):

    x = q_utility(h, c, mass, cap)

    z+=x

    print(f"""

    For T_initial = {h} , T_final= {c} , Mass = {mass} , Specific heat = {cap}

    The value of Q is: {x} W""")

print(f"""

    For heating utility total Q_hot_utility: {z*0.001:.1f} kW""")
```

Appendix 8: (Cooling utility python code)

```
import numpy as np

T_initial_hot = np.array([38.4,230.7,175.7,209.4,90.1,114.1,-28.4,77.0])

T_initial_cold = np.array([147.6,234.1,176.6,218.0,91.1,122.4,20.0,250.0])

m = np.array([7.19,119.95,72.06,11.99,21.17,50.37,0.9,0.83])

Cp = np.array([2161,22616,103146,3554,58007,8802,8512,14972])

z = 0

def q_utility(T_h, T_c, mass, spec_heat):

    return mass * spec_heat * (T_c - T_h)

for h, c, mass, cap in zip(T_initial_hot, T_initial_cold, m, Cp):

    x = q_utility(h, c, mass, cap)
```

```
    z+=x
    print(f"""
    For T_initial = {h} , T_final= {c} , Mass = {mass} , Specific heat = {cap}
    The value of Q is: {x} W""")
print(f"""
    For cooling utility Q_cool_utility: {z*0.001:.1f} kW""")
```

Appendix 9 (For all Process integration calculation please refer excel sheet)

(Emailed it to Dr Stephen Houston)